

G04 Gate A

Parcel Delivery Operations

Protection rule

This is a fresh independent assessment. Do not open the matching Review until a genuine saved attempt exists. Do not use prior Gate solutions, tutorial solving, or AI-generated final answers during the attempt.

Source package

Use only G04_GATE_A_SOURCE as the supplied business input. Treat input files as immutable. Create your own src/, sql/, dags/, infra/, output/, quarantine/, logs/ and evidence/ artifacts.

Required integrated build

K1 - PostgreSQL, modeling, constraints, transactions and database safety

Create/verify a PostgreSQL target with explicit schema, keys, constraints and transaction boundaries. Load accepted rows safely; prove one constraint failure/rollback and one grain/modeling decision. Use EXPLAIN or index reasoning only where it materially supports the query.

K2 - File/API/database ingestion contracts, defensive extraction and schema handling

Ingest the immutable dimension/base files plus paginated API pages. Preserve raw evidence, validate schema/types, quarantine malformed business rows, and prove source-to-accepted counts. Do not silently edit bad source rows.

K3 - Incremental state, CDC, idempotency, late data, replay and atomic checkpoints

Apply the changed delta with a deterministic high-water/checkpoint rule and idempotent upsert. Deduplicate API event identity, identify late-event behavior, and prove replay does not double-count. Inject the supplied failure scenario and prove checkpoint/target atomicity.

K4 - ETL/ELT transformations, quality tests, reconciliation and serving contracts

Transform accepted data into a serving result at a stated grain. Implement executable quality checks for required keys/types/relationships/business validity and source-to-target reconciliation. Save pass/fail evidence, not just final output.

K5 - Docker images, containers, Dockerfile, ports, volumes, networks, Compose and secrets

Containerize the pipeline with a reproducible Dockerfile and Compose design. Demonstrate port/volume/network behavior, non-root runtime where appropriate, configuration outside code, and no plaintext secret committed to the repository.

K6 - Airflow DAGs, schedules, retries, params, connections, sensors, backfills and debugging

Author and execute an Airflow 3 workflow with explicit dependencies, schedule/data-interval reasoning, retry/timeout behavior, params/connections/secrets boundary, waiting/sensor behavior where appropriate, and one controlled backfill/reprocessing proof. Diagnose one failure from logs.

K7 - Integrated operations: observability, recovery, reproducibility, evidence and technical defense

From a clean clone/clean workspace, reproduce the run from the README/runbook. Save run metadata, counts, timings/error class, failure/recovery evidence and a concise technical defense connecting design choices to observed controls.

Objective evidence checklist

Area	Evidence required
Input contract	Raw file hashes or equivalent identity, source grain, schema assumptions, pagination rule and run window.
Database	DDL/schema + load evidence + transaction/constraint proof + one validation query.
Ingestion	Accepted/quarantine outputs and source -> accepted reconciliation.
Incremental/replay	Checkpoint before/after, changed-delta result, same-input rerun proof, late/duplicate event handling.
Failure/recovery	Fail the atomic load after validation but before checkpoint commit. Target and checkpoint must remain unchanged.
Transformation/quality	Serving output, executable quality results, grain statement and independent aggregate reconciliation.
Docker	Dockerfile, Compose/config/secrets evidence, build/run/volume/network proof.
Airflow	DAG code, graph/run evidence, logs for success + one failure/retry/backfill/recovery.
Reproducibility	README/runbook plus clean-workspace/clean-clone reproduction.
Defense	2-4 minute own-words explanation of architecture, failure boundary, state rule, quality controls and one trade-off.

Decision rule

G04 PASS is allowed only when K1-K7 are all DEMONSTRATED with saved evidence and no unresolved critical defect remains. A numeric percentage is not a substitute. If one competency is weak, use targeted repair after Review; assign a fresh full Gate B/C only when a full reassessment is justified.